Project 1: BMI Survey

Introduction to Statistics

Technical University of Denmark (DTU)
Department of Applied Mathematics and Computer Science

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Intro, task analysis

In this project i will use Python to analyse a Danish BMI survey with 145 respondents: 72 women and 73 men. The dataset contains height, weight, gender, city size (urbanity), and fast-food consumption in days per year. There are no missing values. The goal is to describe BMI, investigate whether the population median BMI differs from 25, compare women and men, and explore correlations involving BMI, weight, and fast-food consumption.

BMI calculation

The BMI for respondent i is $\mathrm{BMI}_i = \frac{W_i}{(H_i/100)^2}$, where W_i is weight in kilograms and H_i is height in centimetres. BMI is measured in $\mathrm{kg/m^2}$.

Plan

- **a-e:** Here the plan is to describe the data, calculate BMI, make density histograms and a boxplot, and calculate summary statistics for everyone and for women and men separately.
- **f-h:** This part is basically analyse the full sample. Model log-transformed BMI, check the model using a QQ plot, calculate confidence intervals, and test whether mean log-BMI differs from $\log(25)$.
- **i-l:** Here we compare women and men. Fit separate log-BMI models, calculate confidence intervals, perform Welch's two-sample t-test, and discuss whether the confidence intervals support the same conclusion.
- **m:** Lastly but not least we explore correlations. Calculate correlations and make scatter plots for BMI, weight, and fast-food consumption.

Exercises

a) Description of the data

The dataset contains 145 respondents and five variables. Height (centimetres) and weight (kilograms) are quantitative. Gender is categorical, coded 0 for women and 1 for men. Urbanity is an ordered category from 1 to 5 describing city size. Fast-food consumption is represented by estimated days per year, with values corresponding to 8 ordered frequency categories from never to every day. There are no missing values: the sample contains 72 women and 73 men.

b) Density histogram of BMI

BMI was calculated for each respondent using $\mathrm{BMI}_i = \frac{W_i}{(H_i/100)^2}$. The full-sample density histogram is unimodal, with most scores between about 22 and 30 and a longer tail toward high BMI values. The sample mean, 25.248, is above the median, 24.691, consistent with right skew. BMI cannot be negative when weight and height are positive; all observed

values are positive. BMI ranges from 17.577 to 39.520, and the middle half ranges from $Q_1 = 22.590$ to $Q_3 = 27.636$, showing noticeable variation.

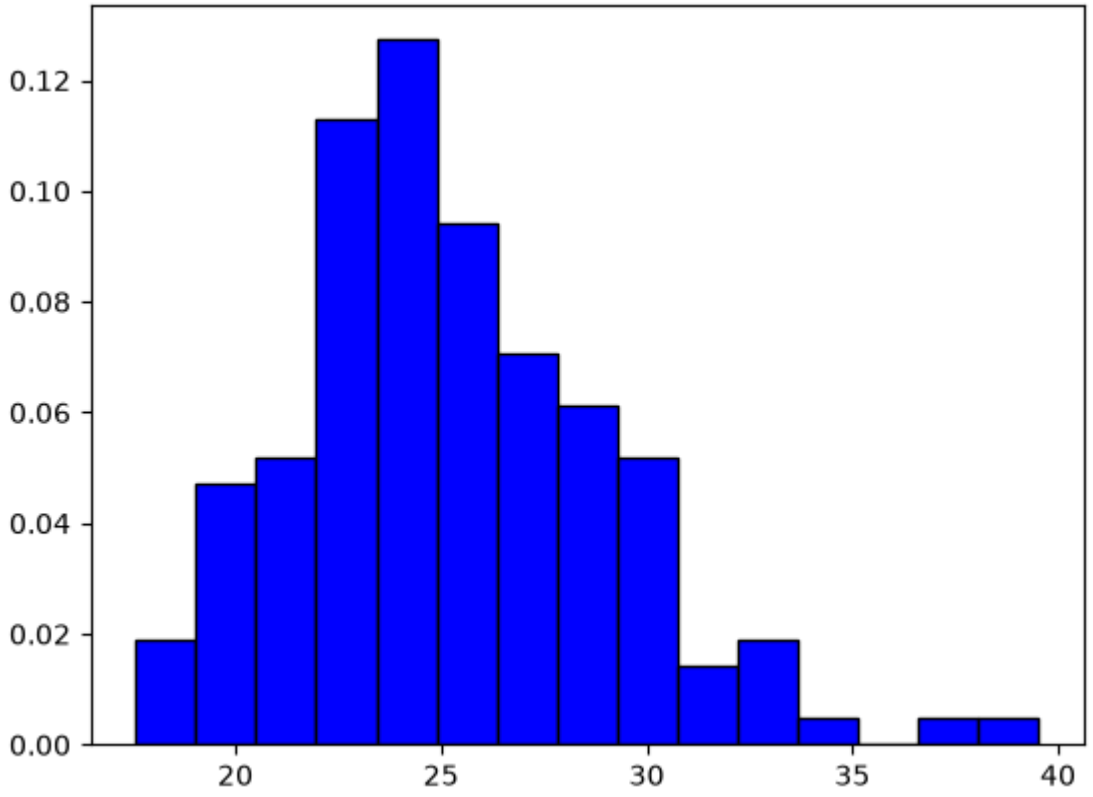


Illustration: Full-sample BMI density histogram.

c) Gender-specific density histograms

The women's histogram is concentrated at somewhat lower BMI values, mostly around 21-25, and has a right tail. The men's histogram is shifted toward higher values, with more observations in the mid-to-upper twenties; it also has a right tail. The distributions overlap substantially, but men appear to have a higher centre. The sample medians are 23.689 for women and 25.726 for men, a difference of about 2.04 BMI units.

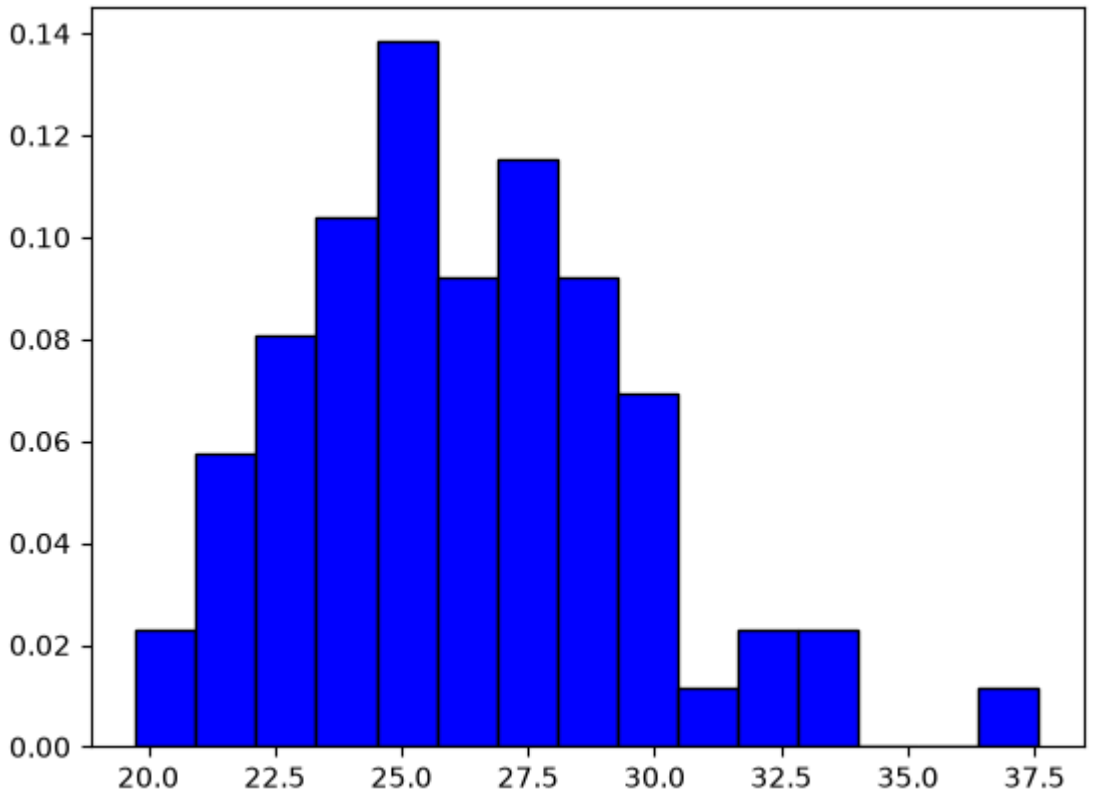


Illustration: BMI density histograms for men.

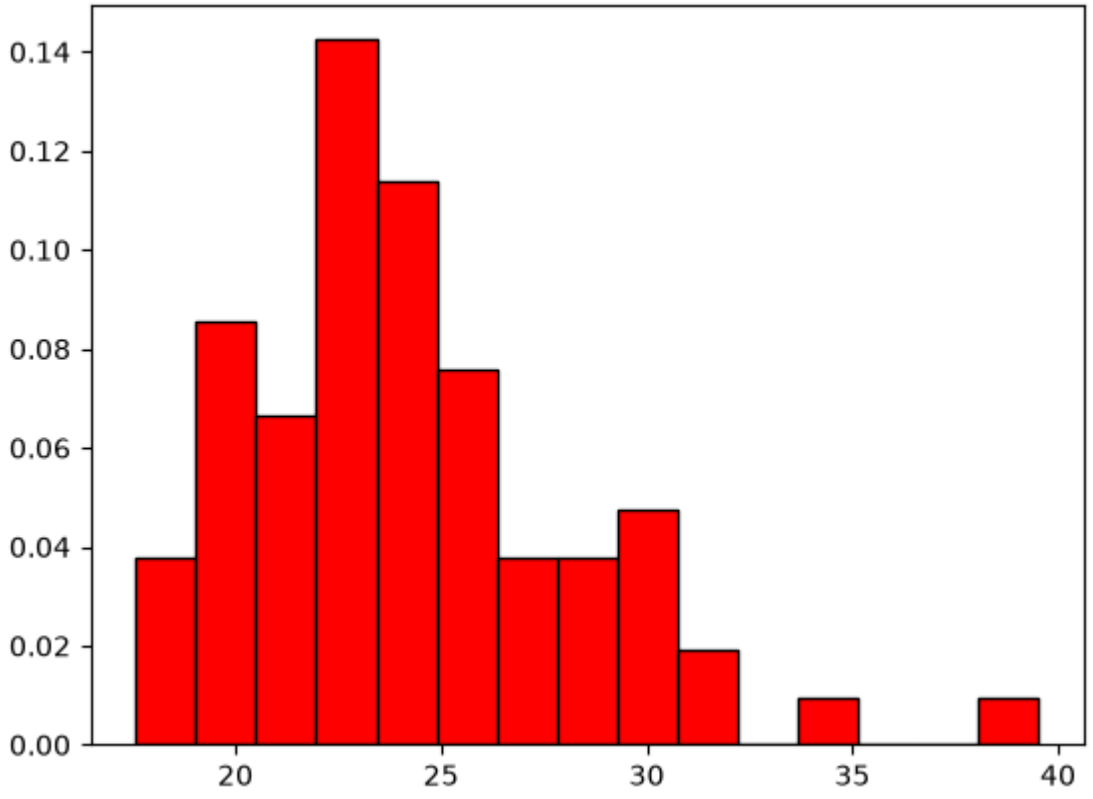


Illustration: BMI density histograms for men.**

d) Boxplot by genders

The boxplot confirms that men's BMI is centred higher: the male median is 25.726 versus 23.689 for women. Both distributions are right-skewed: their upper tails are longer and their means exceed their medians. Women show slightly more spread (IQR $= 5.033$; sample SD $= 4.052$) than men (IQR $= 4.482$; sample SD $= 3.327$). Using the usual $1.5 \times \mathrm{IQR}$ rule, the

boxplot flags two high observations among women (BMI 34.081 and 39.520) and one among men (BMI 37.577). These are statistical outliers under that rule, not necessarily data errors.

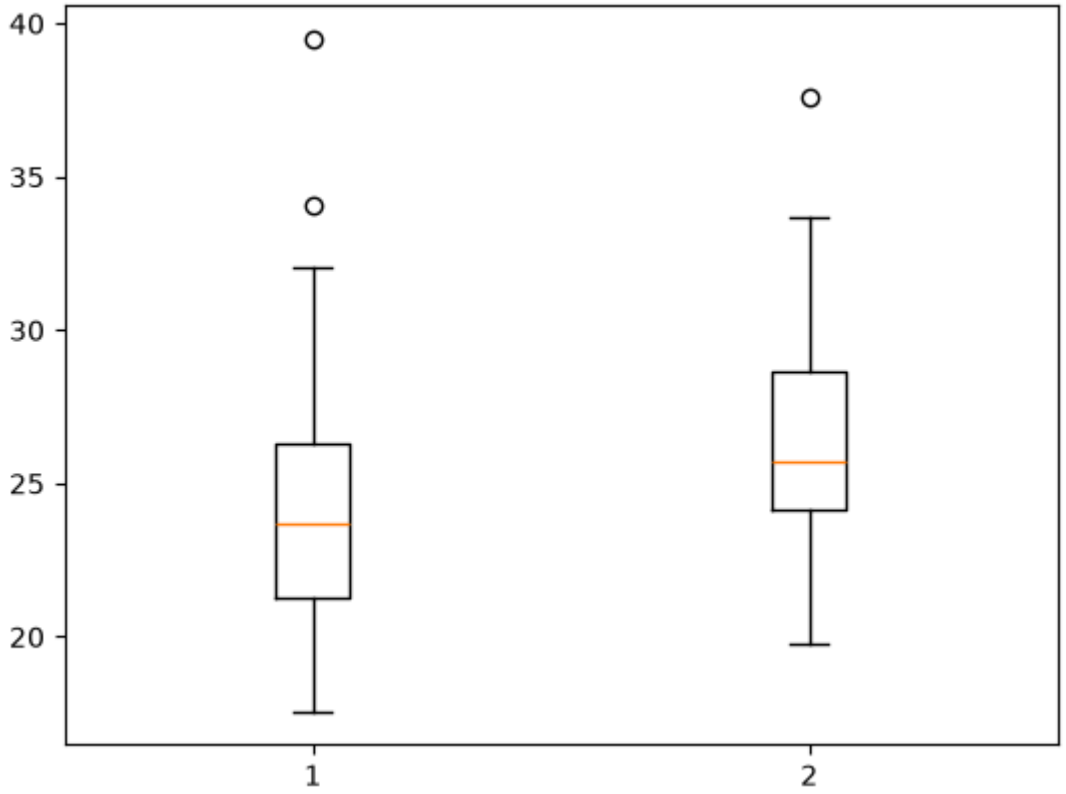


Illustration: BMI boxplot by gender.

e) Summary statistics for BMI

Group	n	Sample mean	Sample variance	Sample SD	Q_1	Median (Q_2)	Q_3
Everyone	145	25.248	14.686	3.832	22.590	24.691	27.636
Women	72	24.216	16.418	4.052	21.259	23.689	26.292
Men	73	26.265	11.069	3.327	24.152	25.726	28.634

BMI is measured in $\mathrm{kg/m^2}$; variance is measured in $(\mathrm{kg/m^2})^2$. Compared with the boxplot, the table gives exact sample sizes, means, variances, standard deviations, and quartiles. The boxplot shows the median, quartiles, and potential outliers visually, but does not show the mean or give these values exactly.

f-h: Full-sample analysis of log-BMI

f) Model and validation

Let $Y_i = \log(\mathrm{BMI}_i)$ for respondent i. We model the full sample, without separating genders, as independent observations from a normal distribution: $Y_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$. For $n = 145$, the estimates are $\bar{Y} = 3.2176$ and $S = 0.1489$.

The QQ plot is close to a straight line through most of the data, so normality of log-BMI looks reasonable. The largest observations depart upward from the line, indicating a small upper-tail deviation. With $n = 145$, the central limit theorem also supports using the t-based methods for the sample mean, since its sampling distribution is approximately normal.

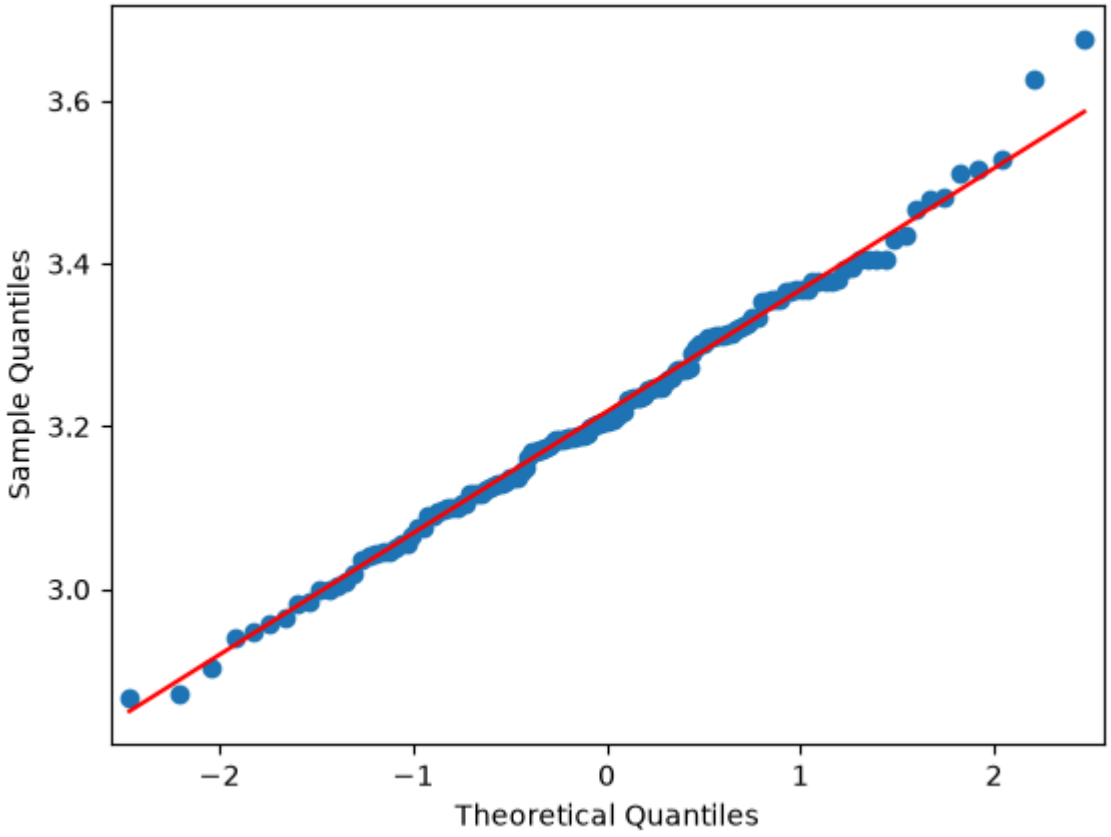


Illustration: Normal QQ plot of log-BMI.

g) Confidence intervals

For the population mean of log-BMI, the 95% confidence interval is $\bar{Y} \pm t_{0.975,n-1}\dfrac{S}{\sqrt{n}}$.

Using $n = 145$, $\bar{Y} = 3.2176$, $S = 0.1489$, and $t_{0.975,144} = 1.9766$, the interval is $[3.1932,\ 3.2421]$. Under the log-normal model, exponentiating the endpoints gives a 95% confidence interval for the population median BMI: $[e^{3.1932}, e^{3.2421}] = [24.366,\ 25.587]\ \mathrm{kg/m^2}$.

The model-based point estimate of the median BMI is $e^{\bar{Y}} = 24.969\ \mathrm{kg/m^2}$.

h) One-sample hypothesis test

We test $H_0 : \mu = \log(25)$ against $H_1 : \mu \neq \log(25)$ at significance level $\alpha = 0.05$. The test statistic is $T = \dfrac{\bar{Y} - \log(25)}{S/\sqrt{n}}$, which follows a t distribution with $n - 1 = 144$ degrees of freedom under H_0.

The python notebook gives $t_{\text{obs}} = -0.0999$ and the two-sided p-value $= 0.9206$. Since $p > 0.05$, we do not reject H_0. The data provide no evidence that the population median BMI differs from 25. In particular, we cannot conclude that more than half of the population has BMI above 25 (is overweight by this cutoff).

i–l: Comparing women and men

i) Separate models and validation

For each gender group $g \in \{W, M\}$, let $Y_{ig} = \log(\text{BMI}_{ig})$. We fit separate normal models, assuming independent observations within each group: $Y_{ig} \overset{\text{iid}}{\sim} N(\mu_g, \sigma_g^2)$. The estimates are:

- Women: $n_W = 72$, $\widehat{\mu}_W = \bar{Y}_W = 3.174097$, and $S_W = 0.159888$.
- Men: $n_M = 73$, $\widehat{\mu}_M = \bar{Y}_M = 3.260588$, and $S_M = 0.123911$.

The separate QQ plots are close to straight lines through most of the observations, supporting the normal model for log-BMI. Both have a small upward departure at the largest values, so the upper tails show mild deviations.

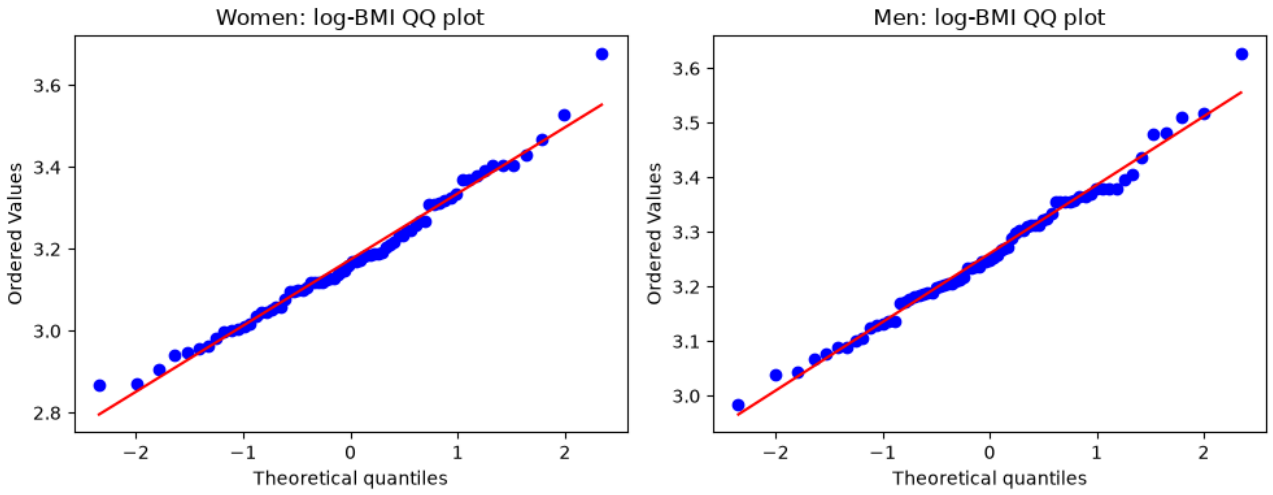


Illustration: QQ plots of log-BMI for women and men.

j) Confidence intervals

For each group, the 95% CI for the mean log-BMI is $\bar{Y}_g \pm t_{0.975, n_g - 1} S_g / \sqrt{n_g}$. Exponentiating the endpoints gives the CI for the median BMI under the log-normal model:

Group	95% **CI for mean log-BMI**	95% **CI for median BMI** (kg/m^2)
Women	$[3.136525,\ 3.211669]$	$[23.023723,\ 24.820469]$
Men	$[3.231677,\ 3.289498]$	$[25.322087,\ 26.829402]$

Hereby the women's median-BMI interval agrees with the notebook's result: $[23.023723,\ 24.820469]$.

k) Welch two-sample t-test

We test $H_0 : \mu_W = \mu_M$ against $H_1 : \mu_W \neq \mu_M$ at $\alpha = 0.05$. The test statistic is

$$T = \frac{\bar{Y}_W - \bar{Y}_M}{\sqrt{S_W^2/n_W + S_M^2/n_M}}, \qquad T \overset{H_0}{\approx} t_\nu,$$

where ν is the Welch–Satterthwaite degrees of freedom. Here, $\nu = 133.7501$, $t_{\text{obs}} = -3.6375$, and the two-sided p-value is 0.000392. Since $p < 0.05$, we reject H_0. The data provide evidence that the mean log-BMI differs by gender; the sample estimates indicate a higher mean log-BMI, and hence a higher median BMI, for men.

l) Comparing the confidence intervals with the test

The group CIs do not overlap: the women's upper limit for mean log-BMI, 3.211669, is below the men's lower limit, 3.231677. The exponentiated median-BMI CIs are also separate. This agrees with the Welch test's conclusion that the groups differ. Comparing separate CIs is an informal check; the Welch test directly tests the difference between the group means.

m) Correlations

The Pearson correlations are:

Variables	Correlation r	Interpretation
BMI and weight	0.8283	Strong positive linear association.
BMI and fast-food days	0.1532	Weak positive linear association.
Weight and fast-food days	0.2793	Weak positive linear association.

The BMI–weight scatterplot shows a clear positive trend. The plots involving fast-food frequency show much weaker patterns; one very large reported frequency (365 days per year) stands apart and may influence the correlations. These are descriptive associations and do not show that fast-food consumption causes changes in BMI or weight.

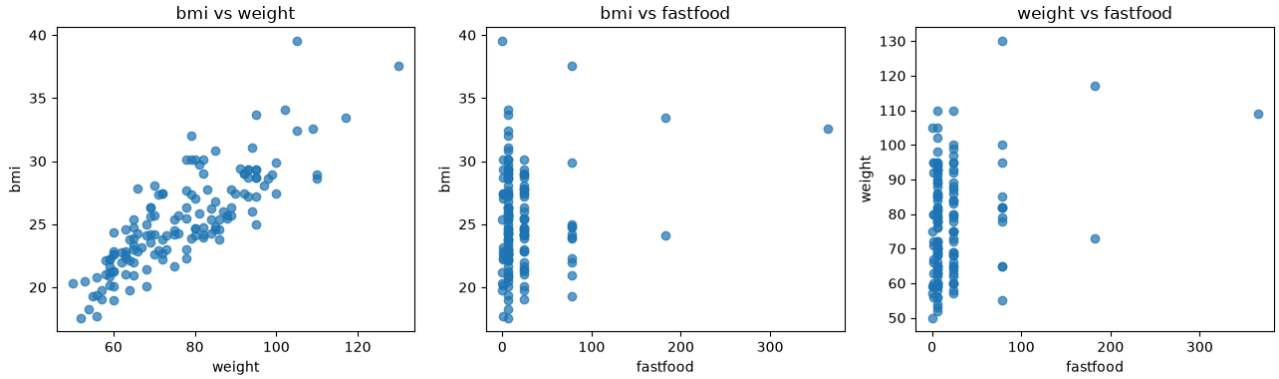


Illustration: Pairwise scatterplots of BMI, weight, and fast-food frequency.